

Sam Park

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Projects

Drumbeat Atlas, Drumbeat Labs

2026

- Architected a social media collection and analysis pipeline on Cloudflare (D1, Workers, R2) with scheduled GitHub Actions fetches and a web management UI, assembling a 37,000-post corpus across seven campaign topics
- Enriched posts with local Whisper transcription and frame OCR, 768-dimension embeddings, PCA, and Leiden clustering, plus transformer and lexicon sentiment scoring of caption, audio, and on-screen text
- Probed the embedding space for five LLM-labeled categories with logistic regression reaching held-out AUC 0.80 to 0.91, then estimated each category's engagement association across 14,000 posts in two campaigns
- Found views are driven by follower count and post age while engagement is associated with semantic content
- Diagnosed a recency and relevance bias in the Scrape Creators API and scoped temporal claims accordingly

Drumbeat Viewer, Drumbeat Labs

2026

- Built an in-browser interface for Drumbeat Atlas exports of post text embeddings, sentiment analysis, and metadata
- Renders the UMAP embedding projection colored by any field via DuckDB-wasm + Mosaic + vgplot
- Links density distribution plots that crossfilter the map in real time, with interval brushes that highlight the selection
- A heatmap strip reports the live deviation from corpus mean and correlations with the colored variable over the current selection

BEDbase + gtars, University of Virginia

2026

- Built an updated web interface for bedbase.org, incorporating an interactive UMAP embedding viewer
- Live-rendered Observable Plot visualizations replace our precomputed ones from ggplot2
- Drove the porting of the lab's algorithms into Rust + WASM bindings to facilitate these features

Genomic Regions, University of Virginia

2026

- Built an interactive multi-panel explorer for Region2Vec genomic region embeddings incorporating linked UMAPs, chromosome distributions, and region co-occurrence plots
- Explored spatial relationships between region embedding tokens as a provisional grammar of regulatory genomics

Experience

Bioinformatician, University of Virginia

2024 – Current

- Built interactive user and data visualization interfaces for tools published by Sheffield Lab
- Developed Refget SCOM interface to visualize comparisons between genomic sequence collections via Vega-Lite
- Ran statistical analyses (stratified GWAS + pQTL colocalization) to identify credible signals for sex differences in lung function
- Investigated markers of abnormal lobe function and transplant rejection using single-cell gene expression in human lung
- Engineered a single-cell RNAseq pipeline for integration with agentic AI, incorporating a canonical config source, machine-readable output summaries, and pipe-friendly I/O
- Contributed to pilot study exploring genetic sources of biological age gap within human proteome and transcriptome (biological age prediction + GWAS)

Automation Engineer, Merck & Co. (Contract)

2022 – 2025

- Helped lead automation community project incorporating R Shiny, plumber API, AWS, and PI Web API to automate manual continuous historian report writing process, eliminating 500+ manually-written reports per year
- Developed initial proof of concept for continuous historian report tool that served as foundation for codebase
- Worked on automated data pipelines that use R Markdown and PI Web API to manage factory data on AWS S3 and RDS

Skills

Languages & Data: R, Python, JavaScript, TypeScript, SQL

Statistics & Experimentation: causal inference, permutation testing, effect estimation, linear regression

Machine Learning: text embeddings, supervised classification and probing, PCA, UMAP, clustering, random forest

Data Visualization: Vega-Lite, Mosaic/vgplot, Observable Plot, D3, ggplot2

Infrastructure: React, React Router, Zustand, FastAPI, Cloudflare, GitHub Actions, DuckDB

Education

University of Virginia

Master of Science (Aug 2026). Program: Data Science

Bachelor of Science (May 2022). Major: Biomedical Engineering | Minor: Computer Science